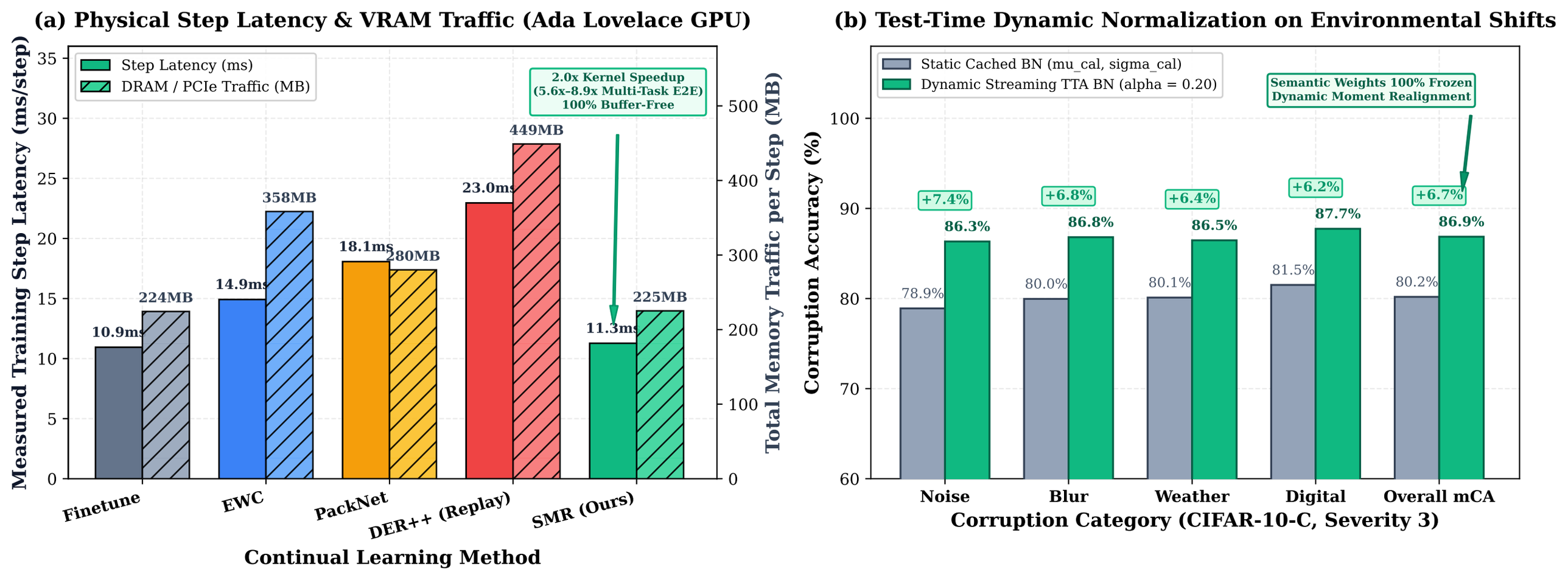
(a) Physical Step Latency & VRAM Traffic (Ada Lovelace GPU)
Step Latency (ms)
DRAM / PCIe Traffic (MB)
2.0x Kernel Speedup
(5.6x–8.9x Multi-Task E2E)
100% Buffer-Free
10.9ms
224MB
14.9ms
358MB
18.1ms
280MB
23.0ms
449MB
11.3ms
225MB
Finetune
EWC
PackNet
DER++ (Replay)
SMR (Ours)
Measured Training Step Latency (ms/step)
Total Memory Traffic per Step (MB)
Continual Learning Method
(b) Test-Time Dynamic Normalization on Environmental Shifts
Static Cached BN (mu_cal, sigma_cal)
Dynamic Streaming TTA BN (alpha = 0.20)
Semantic Weights 100% Frozen
Dynamic Moment Realignment
+7.4%
86.3%
78.9%
+6.8%
86.8%
80.0%
+6.4%
86.5%
80.1%
+6.2%
87.7%
81.5%
+6.7%
86.9%
80.2%
Corruption Accuracy (%)
Noise
Blur
Weather
Digital
Overall mCA
Corruption Category (CIFAR-10-C, Severity 3)